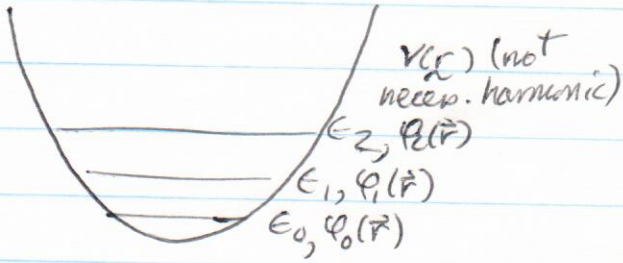
follows Baym Ch. 11

Second Quantization: Intro. to Field Theory

Manipulating permutation operators & Slater determinants is cumbersome for $N > 2$. We develop a special "second quantized" notation to handle much of the bookkeeping. This will be the formalism of quantum field theory, too. Note: This material can't really be "derived" from fundamental principles.

idea: We used (a^+, a) for harmonic oscillator to add and destroy quanta of excitation. Now we let (a^+, a) create, destroy particles

Consider a many Boson system



$\varphi_i(\vec{r})$: single particle eigenstates (appropriately symmetrized products are legitimate MB functions) ← noninteracting

Let $|0\rangle$ = state with no particles present
Let $|n\rangle$ = " " n particles in $\varphi_0(\vec{r})$. ← ONLY one level for now!

Define a_0, a_0^+ by:

particle annihilation operation

$$a_0|n\rangle = \sqrt{n}\,|n-1\rangle$$
$$a_0^+|n\rangle = \sqrt{n+1}\,|n+1\rangle$$

These operators add, subtract a particle to the state $|n\rangle$ — add/subtract one particle to the ground state $\varphi_0(\vec{r})$.

↑ Looks like harmonic oscillator

Required

Commutation relations for Bosons: For an arbitrary $|n\rangle$

$$(a_0 a_0^\dagger - a_0^\dagger a_0)|n\rangle = [(n+1) - n]|n\rangle \Rightarrow [a_0, a_0^\dagger] = 1$$

$= |n\rangle$

acting to right

From the definition, $|n\rangle = \frac{(a_0^\dagger)^n}{\sqrt{n!}} |0\rangle$

$|0\rangle$: "no particle" "vacuum state"

$\frac{(a_0^\dagger)^n}{\sqrt{n!}}$: add n particles to state $\varphi_0(\vec{r})$, starting with no particles "vacuum" state.

As with harmonic oscillator, $(a_0^\dagger)$ acting to left behaves like its HC a_0:

$$\langle n| a_0^\dagger = \sqrt{n}\,\langle n-1|$$

similarly

$$\langle n| a_0 = \sqrt{n+1}\,\langle n+1|$$

$\hat{N} = a_0^\dagger a_0$ measures # of particles in a state φ_0 $\left[\hat{N}|n\rangle = n|n\rangle\right]$.

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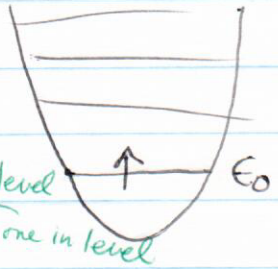
What about Fermions?

Fermions



Consider case for which particles are in lowest level with spin up. Two such states: $|0\rangle$ and $|1\rangle$ by exclusion principle. Define operators to include antisymmetry requirements etc.

Define annihilation & creation ops by:

$$a_0|0\rangle = 0 \qquad a_0|1\rangle = |0\rangle$$

$$a_0^\dagger|0\rangle = 1 \qquad a_0^\dagger|1\rangle = 0 \leftarrow \text{Here's where we tell the formalism that there are fermions!}$$

$$a_0 = \begin{bmatrix} \langle 0|a_0|0\rangle & \langle 0|a_0|1\rangle \\ \langle 1|a_0|0\rangle & \langle 1|a_0|1\rangle \end{bmatrix} \begin{matrix} \langle 0| \\ \langle 1| \end{matrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$a_0^\dagger = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

Commutation relations? They aren't simple

$|n\rangle = |0\rangle$ or $|1\rangle$: $[a_0, a_0^\dagger]|n\rangle = a_0 \begin{cases} 0 & n=1 \\ |1\rangle & n=0 \end{cases} - a_0^\dagger \begin{cases} |0\rangle & n=1 \\ 0 & n=0 \end{cases}$

$$\neq f(n)|n\rangle$$

However ANTIcommutation relations are simple.

Define $\{\hat{\alpha}, \hat{\beta}\} = \hat{\alpha}\hat{\beta} + \hat{\beta}\hat{\alpha}$

Its easy to see that

$$\{a_0, a_0^\dagger\} = \hat{\mathbb{1}}$$ (exes: verify this)

also $a_0^2 = (a_0^\dagger)^2 = 0$, since $\begin{cases} a_0 a_0 |1\rangle = a_0|0\rangle = 0 \\ a_0^\dagger a_0^\dagger |0\rangle = a_0^\dagger |1\rangle = 0 \end{cases}$ etc

$N_0 = a_0^\dagger a_0$ still measures # of (fermion) particles.

Now, consider occupying more than one level P_0.

Bosons: Simple extension. New notation:

$|n_0, n_1\rangle$

↑ # parts in state 0 ↑ # parts in state 1

As you expect,

$$a_0 |n_0, n_1\rangle = \sqrt{n_0}\, |n_0 - 1, n_1\rangle$$

Define as

$$a_0^\dagger |n_0, n_1\rangle = \sqrt{n_0 + 1}\, |n_0 + 1, n_1\rangle$$

$$a_1 |n_0, n_1\rangle = \sqrt{n_1}\, |n_0, n_1 - 1\rangle$$

$$a_1^\dagger |n_0, n_1\rangle = \sqrt{n_1 + 1}\, |n_0, n_1 + 1\rangle$$

Clearly $[a_0, a_0^\dagger] = 1$, $[a_1, a_1^\dagger] = 1$, $[a_0, a_1] = 0$ etc

Arbitrary state can be built:

$$|n_0, n_1\rangle = \frac{(a_1^\dagger)^{n_1}}{\sqrt{n_1!}} \frac{(a_0^\dagger)^{n_0}}{\sqrt{n_0!}} |0, 0\rangle$$

Total number operator $N = a_0^\dagger a_0 + a_1^\dagger a_1$ $\quad N|n_0, n_1\rangle = (n_0+n_1)|n_0,n_1\rangle$

Fermions:

Construction is similar, but require sign change under particle exchange. Antisymmetry leads to anticommutation relation:

$$\{a_0, a_0^\dagger\} = 1 \quad \{a_1, a_1^\dagger\} = 1 \quad \{a_0, a_0\} = \{a_1, a_1\} = 0$$
$$\{a_0^\dagger, a_0^\dagger\} = \{a_1^\dagger, a_1^\dagger\} = 0$$

only 1 fermion at most.

also, "0 and 1" operators anticommute:

$$\{a_0, a_1\} = \{a_0^\dagger, a_1^\dagger\} = \{a_0, a_1^\dagger\} = \{a_0^\dagger, a_1\} = 0$$

Do stuff on left page ↓

To complete the definition need $a_0^\dagger|11\rangle = a_0|01\rangle = 0$ and $a_0|11\rangle = -|01\rangle$

arb. state: $|n_0, n_1\rangle = (a_1^\dagger)^{n_1}(a_0^\dagger)^{n_0}|0,0\rangle$

no factorials needed $(0! = 1, 1! = 1)$!

General case for either Bose or Fermi:

state $|n_0, n_1, n_2 \cdots n_{max}\rangle$ etc

$$= \cdots \frac{(a_2^\dagger)^{n_2}}{\sqrt{n_2!}} \frac{(a_1^\dagger)^{n_1}}{\sqrt{n_1!}} \frac{(a_0^\dagger)^{n_0}}{\sqrt{n_0!}} |0\rangle \quad \text{Bose}$$

$$= \cdots (a_2^\dagger)^{n_2}(a_1^\dagger)^{n_1}(a_0^\dagger)^{n_0}|0\rangle \quad \text{fermi}$$

Fermi. With 2 levels have $|00\rangle |10\rangle |01\rangle |11\rangle$, with fixed spin.

Then $a_1^+|00\rangle = |01\rangle$ $a_1^+|10\rangle = |11\rangle$

$a_1^+|01\rangle = a_1^+|11\rangle = 0$

Define: (Jordan-Wigner)

$a_1|00\rangle = a_1|10\rangle = 0$

$a_1|01\rangle = |00\rangle$ $a_1|11\rangle = |10\rangle$

$a_0^+|00\rangle = |10\rangle$ $a_0^+|10\rangle = 0$

$a_0|10\rangle = |00\rangle$ $a_0|00\rangle = 0$

and enforce exchange antisymm

exchange in SQ notation

To complete rules for fermions need to enforce antisymmetry under exchange. We mimic exchange by combinations of a's & a^+'s.

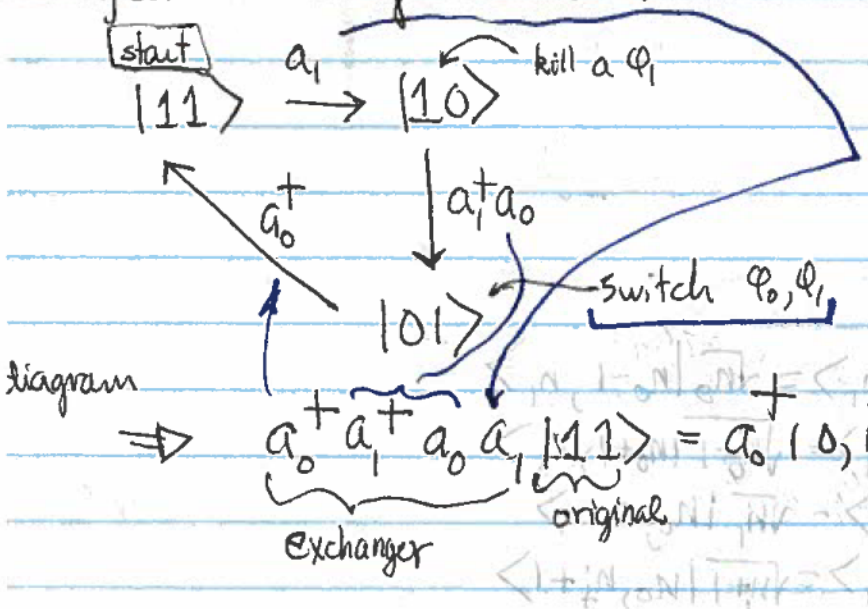


diagram $\Rightarrow$ $a_0^+ a_1^+ a_0 a_1 |11\rangle = a_0^+ |0,1\rangle = -|11\rangle$

$a_0^+ a_1^+ a_0 a_1$: exchanger; $|11\rangle$: original; MUST be (since Fermions)

That is, we must build in the condition that

$a_0^+|01\rangle = -|11\rangle$

or $a_0|11\rangle = -|01\rangle$!

To get proper fermion behavior.

Example: Consider plane waves in a box with periodic BC

$$\varphi_{\vec{p}}(\vec{r}) = \frac{e^{i\vec{p}\cdot\vec{r}}}{\sqrt{V}} \qquad p_x = \frac{2\pi n_x}{L_x} \qquad n_x = 0, \pm 1, \pm 2, \cdots$$

similar p_y, p_z

↑ Single particle unsymm states

a^+_{ps} → add particle with momentum $\vec{p}$, spin s

a_{ps} delete " " " " " "

The a's are localized in momentum space — but suitable linear combination is localized in real space:

Define: $\psi^+_s(\vec{r}) \equiv \sum_{\vec{p}} \frac{e^{-i\vec{p}\cdot\vec{r}}}{\sqrt{V}} a^+_{ps}$ Field operator

The field operator creates a particle at $\vec{r}$ with spin s

proof: The amplitude for finding a particle at $\vec{r}'$ is

$$\langle \vec{r}' | \psi^+_s(\vec{r}) | 0 \rangle = \sum_{\vec{p}} \frac{1}{\sqrt{V}} e^{-i\vec{p}\cdot\vec{r}} \underbrace{\langle \vec{r}' | a^+_{ps} | 0 \rangle}_{\langle \vec{r}' | \vec{p}s \rangle}$$

$$\langle \vec{r}' | \vec{p}s \rangle = e^{i\vec{p}\cdot\vec{r}'} \Big/ \sqrt{V}$$

$\int \langle r'|k\rangle \langle \vec{k}s|\vec{p}s\rangle dk$

$\int e^{ik\cdot r'} \delta(k-p)$

↑ change var. to $\vec{k}$

$$= \frac{1}{V} \sum_{\vec{p}} e^{i\vec{p}\cdot(\vec{r}-\vec{r}')} = \delta(\vec{r}-\vec{r}')$$

Similarly $\psi_s(\vec{r}) = \left[\psi_s^\dagger(\vec{r})\right]^\dagger = \sum_{\vec{p}} \frac{1}{\sqrt{V}} e^{i\vec{p}\cdot\vec{r}}\, a_{\vec{p}s}$ removes a particle at $\vec{r}$.

Haven't said whether Boson or Fermion yet.

$$\left\{ \begin{array}{l} \psi_s(\vec{r})\,\psi_{s'}(\vec{r}') \mp \psi_{s'}(\vec{r}')\,\psi_s(\vec{r}) = 0 \\ \psi_s^\dagger(\vec{r})\,\psi_{s'}^\dagger(\vec{r}') \mp \psi_{s'}^\dagger(\vec{r}')\,\psi_s^\dagger(\vec{r}) = 0 \end{array} \right\} \quad \begin{pmatrix} - \text{Boson} \\ + \text{Fermion} \end{pmatrix}$$

(easy proof — write $\psi, \psi^\dagger$ in terms of the a's and use their commutation relation to prove this!)

adding/removing a particle at $\vec{r}$ commutes with adding/removing at $\vec{r}'$

Also:

$$\psi_s(\vec{r})\,\psi_{s'}^\dagger(\vec{r}') \mp \psi_{s'}^\dagger(\vec{r}')\,\psi_s(\vec{r}) = \delta(\vec{r}-\vec{r}')\,\delta_{ss'} \qquad \begin{matrix} - \text{Boson} \\ + \text{Fermion} \end{matrix}$$

Baym

Adding particle at $\vec{r}'$, subtracting at $\vec{r}$ commutes unless do adding/removing at same point. Then, for example if no particle at $\vec{r}$ $\psi^\dagger(r)\psi(r) = 0$ but $\psi(r)\psi^\dagger(r) \neq 0$.

Localization in position:

$$|\vec{r}_1, \vec{r}_2 \cdots, \vec{r}_n\rangle = \frac{1}{\sqrt{n!}}\, \psi^\dagger(\vec{r}_n) \cdots \psi^\dagger(\vec{r}_2)\,\psi^\dagger(\vec{r}_1)|0\rangle$$

↑ state with particles localized at $\vec{r}_1, \vec{r}_2, \cdots \vec{r}_n$.

Note how convenient $\psi, \psi^\dagger$ is for spatially localized states!

Symmetry is Built in

Example: Fermions

Interchange particles at $\vec{r}_1, \vec{r}_2$:

$$|r_1 r_2\rangle = \frac{1}{\sqrt{2}} \psi^\dagger(\vec{r}_2)\psi^\dagger(\vec{r}_1)|0\rangle$$

$$|r_2 r_1\rangle = \frac{1}{\sqrt{2}} \psi^\dagger(\vec{r}_1)\psi^\dagger(\vec{r}_2)|0\rangle = \underset{\text{anti}}{-}\frac{1}{\sqrt{2}} \psi^\dagger(\vec{r}_2)\psi^\dagger(\vec{r}_1)|0\rangle = -|r_1 r_2\rangle \checkmark$$

from commutation relations

This is the whole point of second quantization: it "knows" about MB symmetries

Note that $\psi^\dagger$ adds particles to the right and destroys to the left:

$$\langle r_1 \cdots r_n| = \left[\psi^\dagger(r_n)\psi^\dagger(r_{n-1})\cdots\psi^\dagger(r_1)|0\rangle\right]^\dagger = \frac{}{\sqrt{n!}} \langle 0|\psi(r_1)\cdots\psi(r_n)|\sqrt{n!}$$

"Normalization" matrix element:

$$\langle r_1' r_2' \cdots r_n' | r_1 r_2 \cdots r_n\rangle = \frac{1}{\sqrt{n!}}\frac{1}{\sqrt{n!}} \langle 0|\psi(r_1')\cdots\psi(r_n')\psi^\dagger(r_n)\cdots\psi^\dagger(r_1)|0\rangle$$

Pf: Baym - general case

$$= \frac{\delta_{nn'}}{n!} \sum_P (\pm)^P P\,\delta(r_1 - r_1')\delta(r_2 - r_2')\cdots\delta(r_n - r_n') \quad \left(\begin{matrix}+\text{ Bose}\\ -\text{ Fermi}\end{matrix}\right)$$

↑ Permutation op

Demonstration for $N=2$ particles: Next example is very elementary, but shows how symmetrization arises in the formalism.

Discussion for 2 particles

$\langle x_1, x_2 | r_1, r_2 \rangle$ = "there are particles at x_1, x_2" ; r_1, r_2 = "I put particles at r_1, r_2"

$$\langle x_1, x_2 | r_1, r_2 \rangle = \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}} \langle 0 | \psi(x_1)\psi(x_2)\psi^\dagger(r_2)\psi^\dagger(r_1) | 0 \rangle$$

idea: Keep commuting the ψ's to right or ($\psi^\dagger$'s to left). You're done when get stuff like $\psi(\vec{r})|0\rangle \equiv 0$.

Commutation relations

$$\psi(r)\psi(r') \mp \psi(r')\psi(r) = 0 \qquad \begin{matrix} -\ B \\ +\ F \end{matrix}$$

$$\psi^\dagger(r)\psi^\dagger(r') \mp \psi^\dagger(r')\psi^\dagger(r) = 0$$

$$\psi(r)\psi^\dagger(r') \mp \psi^\dagger(r')\psi(r) = \delta(r-r')$$

lets do Bosons to illustrate:

$$\psi(x_1)\psi(x_2)\psi^\dagger(r_2)\psi^\dagger(r_1) = \psi(x_1)\left[\delta(x_2-r_2) + \psi^\dagger(r_2)\psi(x_2)\right]\psi^\dagger(r_1)$$

$$= \psi(x_1)\delta(x_2-r_2)\psi^\dagger(r_1) + \psi(x_1)\psi^\dagger(r_2)\psi(x_2)\psi^\dagger(r_1)$$ ← use CR to "move" this over

$$= \psi(x_1)\delta(x_2-r_2)\psi^\dagger(r_1) + \psi(x_1)\psi^\dagger(r_2)\left[\delta(x_2-r_1) + \psi^\dagger(r_1)\psi(x_2)\right]$$

$$= \underbrace{\psi(x_1)\delta(x_2-r_2)\psi^\dagger(r_1)}_{①} + \underbrace{\psi(x_1)\psi^\dagger(r_2)\delta(x_2-r_1)}_{②} + \underbrace{\psi(x_1)\psi^\dagger(r_2)\psi^\dagger(r_1)\psi(x_2)}_{③}$$ ③ kill vacuum

$$\therefore \langle x_1, x_2 | r_1, r_2 \rangle = \frac{1}{2}\langle 0 | ① + ② + ③ | 0 \rangle$$

Can't annihilate particle from vacuum state!

$$= \frac{1}{2}\left[\langle x_1 | r_1 \rangle \delta(x_2-r_2) + \langle x_1 | r_2 \rangle \delta(x_2-r_1) + 0\right]$$

$$= \frac{1}{2}\left[\delta(x_1-r_1)\delta(x_2-r_2) + \delta(x_1-r_2)\delta(x_2-r_1)\right]$$

Symmetry is built in

Fermions would have given us a − sign here! (Try it!)

Position representation & symmetrization

$|\Phi\rangle$ = many body state

$\varphi(r_1 r_2 \cdots r_n)$ = unsymmetrized state

$$|\Phi\rangle = \int d^3r_1 \int d^3r_2 \cdots \int d^3r_n \, \varphi(r_1 r_2 \cdots r_n) \, |r_1 r_2 \cdots r_n\rangle$$

Recall: of $|\Phi\rangle = \int dx \varphi(x) |x\rangle$
$\langle x'|\Phi\rangle = \int dx \varphi(x) \langle x'|x\rangle = \varphi(x')$

state

MB
$\Phi(X_1 X_2 \cdots X_n)$ ← wavefcn

★ $\langle X_1 X_2 \cdots X_n | \Phi \rangle$ is properly symmetric even if φ is not!

Example: $n=2$, Bosons

$$\langle X_1 X_2 | \Phi \rangle = \langle X_1 X_2 | \int d^3r_1 \int d^3r_2 \varphi(r_1 r_2) |r_1 r_2\rangle$$

amplitude for finding particle at X_1, X_2

Here is where the symmetrization comes from

$$= \int d^3r_1 \int d^3r_2 \varphi(r_1 r_2) \langle X_1 X_2 | r_1 r_2 \rangle$$

$$= \frac{1}{2} \sum_P (+1)^P P \, \delta(r_1 - X_1)\delta(r_2 - X_2)$$

$$= \frac{1}{2}\left[\delta(r_1 - X_1)\delta(r_2 - X_2) + \delta(r_1 - X_2)\delta(r_2 - X_1)\right]$$

$$= \frac{1}{2}\int d^3r_1 d^3r_2 \, \varphi(r_1 r_2)\left\{\delta(r_1 - X_1)\delta(r_2 - X_2) + \delta(r_1 - X_2)\delta(r_2 - X_1)\right\}$$

$$= \frac{1}{2}\left[\varphi(X_1 X_2) + \varphi(X_2 X_1)\right]$$

Note: Normalization is wrong – happens unless φ was already symmetric

exc: Take properly symmetric function and verify that normalization is OK